arXiv:2311.14752v2 [cond-mat.soft] 26 Sep 2024

Roadmap on machine learning glassy dynamics

Gerhard Jung,[1,2] Rinske M. Alkemade,[3] Victor Bapst,[4] Daniele Coslovich,[5] Laura Filion,[3] François P. Landes,[6] Andrea Liu,[7,8] Francesco Saverio Pezzicoli,[6] Hayato Shiba,[9] Giovanni Volpe,[10] Francesco Zamponi,[11] Ludovic Berthier,[1,12] and Giulio Biroli[11]

[1] *Laboratoire Charles Coulomb (L2C), Université de Montpellier, CNRS, 34095 Montpellier, France*
[2] *Laboratoire Interdisciplinaire de Physique (LIPhy), Université Grenoble Alpes, 38402 Saint-Martin-d'Hères, France*
[3] *Soft Condensed Matter and Biophysics, Debye Institute for Nanomaterials Science, Utrecht University, Utrecht, Netherlands*
[4] *GoogleDeepMind, London, UK*
[5] *Dipartimento di Fisica, Università di Trieste, Strada Costiera 11, 34151, Trieste, Italy*
[6] *Université Paris-Saclay, CNRS, INRIA, Laboratoire Interdisciplinaire des Sciences du Numérique, TAU team, 91190 Gif-sur-Yvette, France*
[7] *Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, PA 19104, USA*
[8] *Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501, USA*
[9] *Graduate School of Information Science, University of Hyogo, Kobe 650-0047, Japan*
[10] *Department of Physics, University of Gothenburg, Origovägen 6B, Gothenburg 41296, Sweden*
[11] *Laboratoire de Physique de l'Ecole Normale Supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université de Paris, F-75005 Paris, France*
[12] *Gulliver, UMR CNRS 7083, ESPCI Paris, PSL Research University, 75005 Paris, France*

(Dated: September 27, 2024)

Unraveling the connections between microscopic structure, emergent physical properties, and slow dynamics has long been a challenge when studying the glass transition. The absence of clear visible structural order in amorphous configurations complicates the identification of the key physical mechanisms underpinning slow dynamics. The difficulty in sampling equilibrated configurations at low temperatures hampers thorough numerical and theoretical investigations. This perspective article explores the potential of machine learning (ML) techniques to face these challenges, building on the algorithms that have revolutionized computer vision and image recognition. We present recent successful ML applications, as well as many open problems for the future, such as transferability and interpretability of ML approaches. We highlight new ideas and directions in which ML could provide breakthroughs to better understand the fundamental mechanisms at play in glass-forming liquids. To foster a collaborative community effort, this article also introduces the "GlassBench" dataset, providing simulation data and benchmarks for both two-dimensional and three-dimensional glass-formers. We propose critical metrics to compare the performance of emerging ML methodologies, in line with benchmarking practices in image and text recognition. The goal of this roadmap is to provide guidelines for the development of ML techniques in systems displaying slow dynamics, while inspiring new directions to improve our theoretical understanding of glassy liquids.

I. INTRODUCTION

When supercooled liquids undergo a glass transition, a dramatic slowdown of transport properties is observed and the resulting material dynamically resembles a crystalline solid — yet one of the main characteristic of glasses is that they maintain their amorphous liquid structure [1]. While glasses and other amorphous materials have been used since prehistoric times, their technological applications have blossomed in recent years [2], including, e.g., metallic glasses for biomedical implants [3] and vapor-deposited organic films used extensively in visual displays [4]. Despite several decades of research involving experiments, theory and computer simulations, many fundamental mechanisms remain to be elucidated, such as macroscopic mechanical properties, highly cooperative stress relaxation in glasses, and the statistical mechanics nature of the glass transition itself [5].

The raise of deep learning in the last decade [6] was initially driven by applications in computer vision, in particular image recognition and feature detection, which soon outperformed traditional techniques [7]. These original breakthroughs are now starting to revolutionize several other areas in technology and science. Our aim here is to address the potential of ML methods to boost research on fundamental aspects of glassy dynamics, in particular the ones that are playing an important role in advancing theories of the glass transition.

One of the main challenges in developing a fundamental microscopic theory of glasses is the absence of any simple and visible structural order. While crystalline defects in the otherwise well-ordered structures are easily detectable, it is an open problem to find analogous structural features in amorphous materials. Over the years, many different proposals for "defects" or locally preferred structures have been proposed and developed [8–11]. This seems to indicate that, even in amorphous configurations, it could be possible to detect the emergence of some kind of short- and medium-range order. However, these identifications usually only apply to specific systems, and they are often only weakly correlated with local dynamical relaxations. There is therefore a clear need for

new and more powerful system-independent ways to systematically find preferred structures in amorphous configurations. This is a challenge for which new ML methods could be a great asset, in particular thanks to the progress in unsupervised learning. Several approaches have been developed recently towards this goal [12–15].

Another long-standing challenge has been understanding and characterizing the fundamental mechanisms underpinning slow and glassy dynamics, which are responsible for the glass transition. To this aim, there has been a substantial effort to identify the microscopic properties that lead to dynamical relaxation. Given a snapshot (an equilibrium configuration), several local properties have been proposed to pinpoint the regions more likely to relax within a window of, say, some fraction of the relaxation time. Examples include the local Debye-Waller factor, eigenvectors of the Hessian of inherent structures, etc. [16, 17]. There is no consensus on what are the best predictors of future dynamics. Moreover, they could change with temperature or be system-specific, according to several theories of the glass transition [18]. Thanks to the advances in numerical simulations of glass-forming liquids we can now produce large datasets of initial configurations and subsequent dynamical trajectories. This provides a natural playground to apply supervised learning techniques in order to identify the local predictors of dynamical relaxation. Several researchers have taken up this challenge and developed ML methods to predict where local relaxations are more likely to take place given an initial snapshot [19–30].

Finally, the ultimate goal of the research efforts devoted to the theory of glassy dynamics would be to combine the solutions of the previous problems to develop an effective theory of the glass transition. Until now, this challenge has been tackled starting from some theoretical assumptions driven by experimental and simulation results [18]. Also in this challenge ML methods can make a difference; they can assist in this quest by providing a complementary identification of the mechanisms inducing relaxation, as first shown in [31].

Clearly, the time is ripe for investigating the ability of the new ML methods to advance our fundamental understanding of glass-forming liquids. In this roadmap, focusing on the three main goals described above, we present respectively in Secs. II, III, IV the recent contributions in this endeavor, discussing the main difficulties ahead and the possible paths to circumvent them. In Fig. 1, we give a visual overview over the different ML concepts that will be discussed within the individual sections. In Sec. V we provide a framework "GlassBench" to enable, encourage and structure a broader community effort to further develop such ML approaches. It consists in (i) a dataset including simulation data for a two-dimensional [30] and a three-dimensional glass-former [25, 32], (ii) benchmarks on different tasks associated to predicting local dynamics from a given initial configuration, (iii) an assessment of the current state of the art. Our purpose is to fuel and organize new developments of advanced ML techniques,

FIG. 1. Visual summary of the roadmap. The individual sections at the center are connected to the big questions in the field of glass physics. They are surrounded by the various machine learning concepts used to answer them.

as done in the field of image and text recognition, as well as generative modelling [33]. Finally, we close this roadmap by discussing in Sec. VI exciting new concepts and directions in ML which have the potential to play a very important role in future research on the theory of glassy dynamics.

II. MACHINE LEARNING LOCALLY FAVORED STRUCTURES

Although glass-forming liquids and glasses lack long-range order, close inspection of their atomic structure reveals particle arrangements that are more regular, symmetric, and of lower (free-)energy than the average. Icosahedral local structures are the best known example of such favorable arrangements and they are found in several metallic alloys, colloidal suspensions as well as in computer models of glassy liquids [8]. Such locally favored structures (LFS), distinct from the bulk of the particle arrangements and yet incompatible with crystalline order, are also key ingredients of some theoretical approaches to glass formation [18, 34].

Despite the importance of structural analysis in glassy materials [8, 9], there is at present no generally accepted operational definition of LFS. Standard approaches, such as Voronoi tessellation [35], topological cluster classifi-

cation [36] and other related methods [37, 38], provide a detailed classification of the possible local geometric arrangements. These methods may indicate what are the most frequent or most stable local arrangements, but they are sensitive to thermal fluctuations and tend to provide a too fine-grained classification, which is difficult to exploit in a theoretical setting. Bond-order parameters (BOP) [39] provide yet another route to characterize the local structure of dense particle systems [9]. While this approach offers in principle a systematic description of the local arrangements, the choice of the relevant BOP has been traditionally guided by physical intuition [9], which requires specific and system-dependent a priori knowledge about the relevant symmetries of the local arrangements.

Unsupervised learning methods offer natural system-independent ways to tackle the above issues [40–42]. Along with automated identification of phase transitions [43–45], characterization of the properties of complex materials from high-dimensional datasets is indeed one of the key applications of unsupervised learning in condensed matter physics [41]. The general idea is to first characterize a faithful, high-dimensional representation of the particle local environment based for instance on a systematic bond-order expansion of the local density [46] (see Ref. [47] for a recent review on structural descriptors). Unsupervised ML methods are then used to identify a small number of collective coordinates, $\tilde{X}_i$, that account for the relevant fluctuations of the local structure, thereby reducing the dimensionality of the descriptors. Dimensionality reduction techniques range from simple principal component analysis (PCA), or its kernel variant, to more sophisticated statistical learning methods, like neural network auto-encoders (AE) [40]. These methods may be combined in the future with more advanced approaches, such as self-supervised learning or pre-training (see Sec. VI B), possibly exploiting the intrinsic symmetries of the system [48]. Clustering methods can be finally applied to this reduced structural representation of the material structure, to pinpoint its heterogeneity [41].

The studies highlighted in Ref. [41] focus mostly on ordered materials or disordered systems with covalent or hydrogen bonding, such as amorphous carbon [49] or liquid water [50], where the preferred geometrical order is easy to identify thanks to low coordination numbers. Dense amorphous systems are characterized instead by close-packed arrangements, which provide a challenging benchmark for this kind of structural analysis. In a series of recent papers [12, 13, 51–53], dimensionality reduction and clustering have been applied to models of closed-packed glass-forming liquids. In particular, Boattini *et al.* [12] have used BOP, Gaussian mixture models and a neural network AE to study glassy binary mixtures. Their work revealed a significant structural heterogeneity, suggesting that in these systems, we can distinguish fluctuating regions displaying two different types of local disorder. These regions display a connection with the temporal fluctuations of dynamics, which will be further discussed in Sec. III.

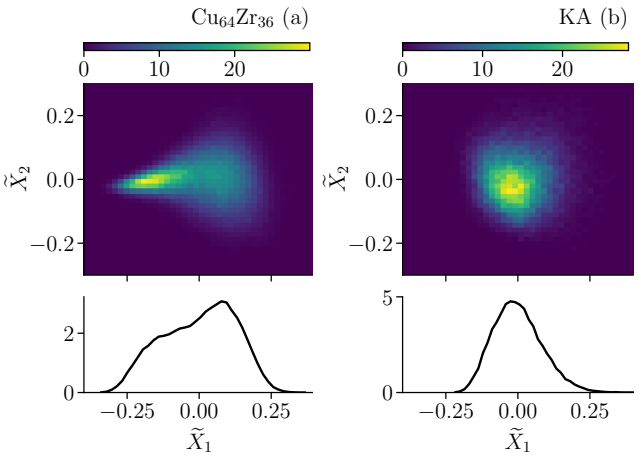

FIG. 2. Principal component analysis maps of the first two principal component projections $\tilde{X}_1$ and $\tilde{X}_2$ of the smooth bond-order parameter around (a) Cu atoms in glassy $Cu_{64}Zr_{36}$ and (b) around small particles in the Kob-Andersen binary mixture around its mode-coupling temperature [51]. Note the contrast between the bimodal structure in (a) due to icosahedral local structures [51] and the homogeneous distribution of the projections in (b). The bottom panels show the marginal distribution of the first principal component projection.

A related study by Paret *et al.* [13] addressed the issue of clustering of local structural arrangements using a different, information-theoretic approach. At a qualitative level, the results of Refs. [12] and [13] appear consistent with one another. However, a more recent investigation [51] revealed a significant system-dependence of structural heterogeneity in glassy liquids. The gist of these findings is illustrated in Fig. 2, which shows representative PCA maps obtained from a smooth bond-order (SBO) descriptor (see supplemental material (SM) for technical details [54]). The distribution of the first two principal components is bimodal for an embedded-atom model of $Cu_{64}Zr_{36}$, which has a well-defined icosahedral LFS, while it is less heterogeneous for the canonical Kob-Andersen (KA) mixture (see also Sec. V), whose local arrangements display clearly different geometrical states. While these differences question the universality of the concept of LFS, the first few principal component projections always correlate with physically-motivated structural measures [51]. We expect that more information could be harvested by looking at chemically-resolved descriptors [55] and on larger length scales (medium range order). Moreover, computing the intrinsic dimension of structural datasets [45, 56] may provide additional insight into the nature of structural order and its system dependence.

One of the striking observations of Ref. [51] is that neural network AE and PCA yield identical reductions of the BOP descriptors. This suggests that either the local